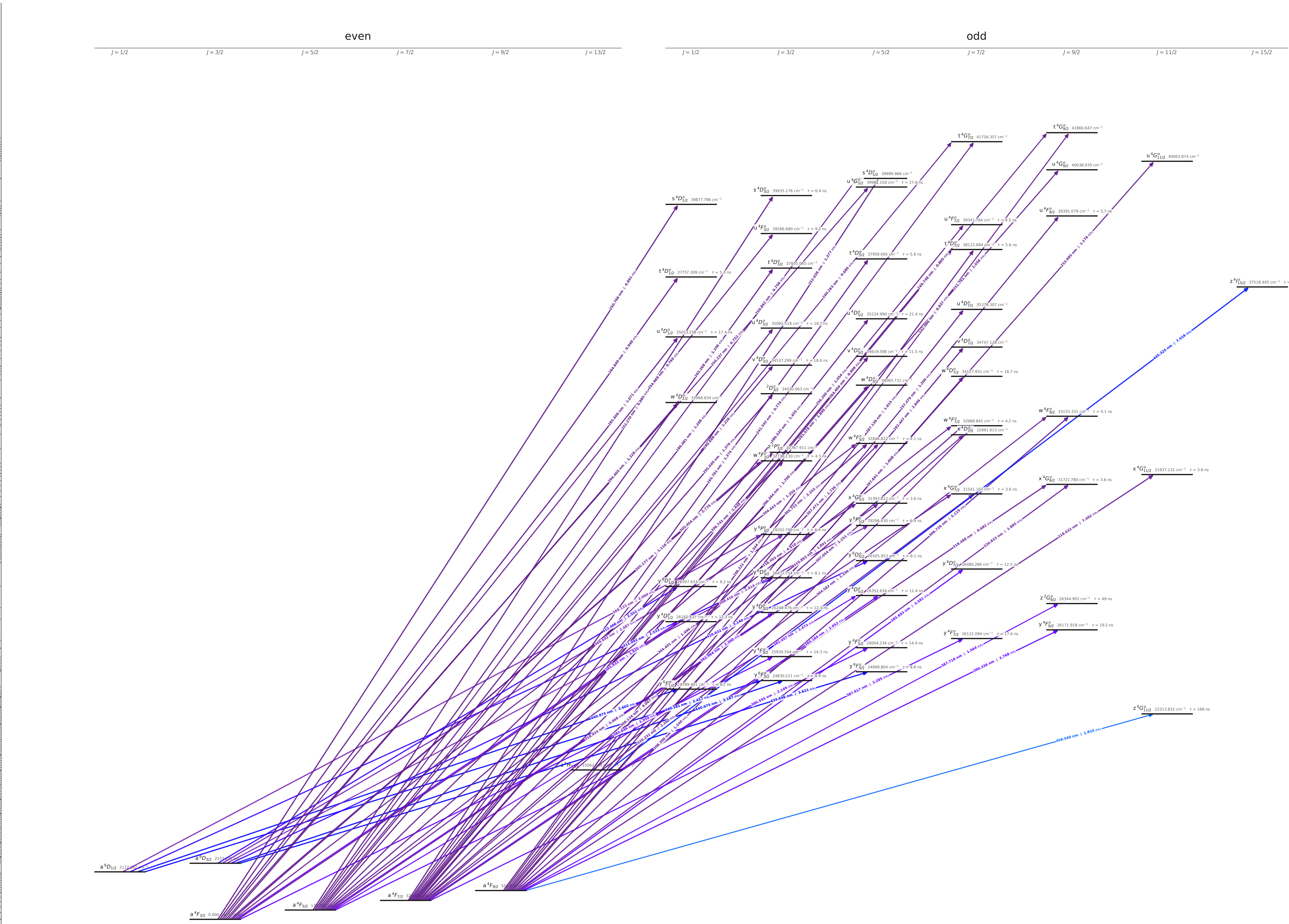


V^+ ionisation limit $54411.67 \text{ cm}^{-1} = 6.74619 \text{ eV} (183.784 \text{ nm})$

Arrow label: vacuum wavelength | reduced dipole matrix element $|U||er||J\rangle$ in $e a_0$
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
 Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: NIST energy and measured lifetime (where one exists).



Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J \text{er} J'\rangle $ (ea ₀)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$a^4H_{13/2} - z^4a_{13/2}^o$	445.3255	445.2005	673.1985	7.516	8.100e+07	12.94	100	laser-cooling

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST value. Γ : natural linewidth of the upper level, $1/(2\pi\tau)$.

ν_{level}	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)										
$yF_{3/2}$	99(30)		F=1: -1112.98	F=2: -915.12	F=3: -618.32	F=4: -222.60	F=5: +272.06	F=6: +865.65					
$yF_{5/2}^0$	15(30)		F=1: -168.63	F=2: -138.65	F=3: -93.69	F=4: -33.73	F=5: +41.22	F=6: +131.16					
$yF_{5/2}^1$	540(60)		F=1: -3642.48	F=3: -2023.60	F=4: +134.91	F=5: +2833.04							
$yF_{5/2}^2$	390(60)		F=1: -4384.46	F=2: -3605.00	F=3: -2435.81	F=4: -876.89	F=5: +1071.76	F=6: +3410.14					
$xG_{3/2}$	603(30)		F=1: -6779.06	F=2: -5573.89	F=3: -3766.14	F=4: -1355.81	F=5: +1657.10	F=6: +5272.60					
$xG_{5/2}$	291(30)		F=0: -4580.08	F=1: -4289.28	F=2: -3707.69	F=3: -2835.29	F=4: -1672.09	F=5: -218.10	F=6: +1526.69	F=7: +3562.29			
$xG_{5/2}^1$	159(30)		F=0: -3058.63	F=2: -2740.85	F=3: -2264.18	F=4: -1628.62	F=5: -834.17	F=6: +119.17	F=7: +1231.40	F=8: +2502.52			
$xG_{5/2}^2$	81(30)		F=2: -1841.48	F=3: -1598.64	F=4: -1274.87	F=5: -870.15	F=6: -384.48	F=7: +182.12	F=8: +829.68	F=9: +1558.17			
$wF_{3/2}^1$	258(30)		F=0: -2900.50	F=2: -2384.85	F=3: -1611.39	F=4: -580.10	F=5: +709.01	F=6: +2255.94					
$wF_{3/2}^2$	108(30)		F=0: -1699.82	F=1: -1591.89	F=2: -1376.04	F=3: -1052.27	F=4: -620.57	F=5: -80.94	F=6: +566.61	F=7: +1322.08			
$zF_{3/2}^1$	183(30)		F=4: -5440.47	F=5: -4526.11	F=6: -3428.87	F=7: -2148.76	F=8: -685.77	F=9: +960.08	F=10: +2788.81	F=11: +4800.42			
$zD_{3/2}^1$	213(30)		F=2: -1436.76	F=3: -798.20	F=4: +53.21	F=5: +1117.48							
$zD_{3/2}^2$	144(30)		F=0: -2266.43	F=1: -2122.53	F=2: -1834.72	F=3: -1403.03	F=4: -827.42	F=5: -107.92	F=6: +755.48	F=7: +1762.78			
$yF_{5/2}^3$	285(30)		F=1: -5482.46	F=2: -4912.85	F=3: -4058.44	F=4: -2919.23	F=5: -1495.22	F=6: +213.60	F=7: +2207.22	F=8: +4485.65			

Measured A, B constants; sources in the data file